

RAJ KHATRI

Data Analytics & Applied AI · Uttarakhand, India

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PROFESSIONAL SUMMARY

Final-year Industrial & Production Engineering student working in data analysis and manufacturing analytics. At Tata Motors, traced a live Body-in-White production defect to a supplier-side dimensional cause using regression-based trend analysis, confirmed it by direct measurement on the shop floor, and designed a containment procedure that needed no line stoppage. Currently a part-time backend developer building the PostgreSQL data layer for a live K-12 ed-tech platform. Strongest in Python, SQL, and applied statistics.

EXPERIENCE

Tata Motors — Industrial Trainee, Data Analytics

3 Jul 2026 – 3 Aug 2026

Project: "Predictive Quality Intelligence — A Data-Driven Framework for BIW Root-Cause Diagnosis"

Pantnagar, Uttarakhand · On-site

- Identified a supplier-side dimensional defect as the root cause of a recurring Body-in-White quality issue, ruling out on-line equipment wear.
- Reached the finding by applying linear regression with R^2 confidence scoring across ~300 inspection data points, then validating it through direct physical measurement on the shop floor.
- Designed a containment procedure deployable without production stoppage or vendor dependency.
- Built an offline, interactive HTML/JS diagnostic dashboard tracking defect trends across 10 monitored quality metrics.
- Authored a 20+ section technical report benchmarking the findings against DFMEA, Six Sigma, and peer-reviewed manufacturing literature.

Voldebug Innovations Pvt. Ltd. — Backend Developer (Part-time)

15 Aug 2026 – Present

Remote

- Migrating a live K-12 AI-literacy platform from browser-local persistence to a Supabase (PostgreSQL) backend — schema design, row-level security policies, and role-based authentication for student, teacher, and principal accounts.
- Implementing server-side APIs in Next.js (App Router) for XP/badge progression, class-code enrollment, and teacher analytics, replacing client-only state with a shared data layer.
- Establishing database migrations, staging/production environment separation, and a repeatable deployment workflow on Vercel — voldebuglearning.vercel.app.
- Authored the phased Supabase + Next.js migration plan now being implemented.

Voldebug Innovations Pvt. Ltd. — Intern

1 Jul 2026 – 31 Jul 2026

Remote

- Built the platform's first working version — a gamified AI-literacy SPA spanning student, teacher, and principal roles, with a curriculum-linked prompt library and an XP/badge/streak progression system.
- Developed AI-assisted lesson-planning tools and a principal-facing analytics dashboard; implemented Hindi/English bilingual support and downloadable student certificates.
- Diagnosed and fixed browser back-navigation bugs using hash-based query parameters, and resolved orphaned timer state affecting session tracking.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL / MySQL), JavaScript, Bash

Data & Analytics: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter, Excel (Advanced / VBA), Power BI

Backend & Tools: Flask, SQLAlchemy, Supabase, Next.js, Streamlit, Git/GitHub, Vercel, VS Code

Methods: Statistical modeling & hypothesis testing, root-cause analysis, DFMEA, Six Sigma, exploratory data analysis, dashboard development

EDUCATION

B.Tech, Industrial & Production Engineering

Aug 2023 – Jun 2027

College of Technology, G.B. Pant University of Agriculture & Technology, Pantnagar

Pantnagar, Uttarakhand

Senior Secondary (XII), CBSE — Maheshwari Public School, Kota · 75.2%

2022

Secondary (X), CBSE — Peace Public School, Gopeshwar · 97.2%

2020

CERTIFICATIONS

Forge Job Simulations — J.P. Morgan (Quantitative Research) · Deloitte Australia (Data Analytics)